Q3. Binary outcome: marginal and random-effects models

To analyze the longitudinal binary outcome, two complementary modeling approaches were considered: a marginal model based on generalized estimating equations (GEE) and a subject-specific model based on a generalized linear mixed model (GLMM). Both approaches account for within-subject correlation but differ fundamentally in their interpretation and treatment of individual heterogeneity.

Marginal model (GEE)

The marginal model was specified as a logistic regression including linear and quadratic effects of time and baseline covariates (age, sex, BMI, ADL, amyloid PET, and tau PET). Within-subject correlation was modeled using a working correlation structure selected via the QIC criterion. Although differences in QIC across independence, exchangeable, and AR(1) structures were small, the exchangeable structure yielded the lowest QIC and was therefore retained.

Let $Y_{ij} \in \{0,1\}$be the binary outcome for subject $i$at visit j, measured at time t_{ij}. Let $\mathbf{x}_i$be the vector of baseline covariates (e.g., age, sex, BMI, ADL, amyloid PET, tau PET).

Mean model (population-averaged)

$$\begin{aligned} \mu_{ij} &= \mathbb{E}(Y_{ij} \mid t_{ij}, \mathbf{x}_i), \\ \operatorname{logit}(\mu_{ij}) &= \beta_0 + \beta_1 t_{ij} + \beta_2 t_{ij}^2 + \mathbf{x}_i^\top \boldsymbol{\beta}. \end{aligned}$$

Working correlation (exchangeable)

Define the working covariance of $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{in_i})^\top$as

$$\operatorname{Var}(\mathbf{Y}_i) = \mathbf{A}_i^{1/2}\, \mathbf{R}(\alpha)\, \mathbf{A}_i^{1/2},$$

where $\mathbf{A}_i = \operatorname{diag}(\mu_{ij}(1-\mu_{ij}))$, and for an exchangeable structure:

$$\mathbf{R}(\alpha)_{jk} = \begin{cases} 1 & j = k, \\ \alpha & j \neq k. \end{cases}$$

(Estimation is via the standard GEE score equations; inference uses robust “sandwich” SEs.)

The GEE results showed a pronounced non-linear time effect. The linear time coefficient was negative and the quadratic term positive, indicating that the population-averaged probability of the event initially decreases and subsequently increases over follow-up. Most baseline covariates were not statistically significant in the marginal model, suggesting that, after adjustment for time, these variables have limited influence on the population-average risk. As expected, GEE estimates correspond to population-averaged odds ratios describing how the mean response evolves across the population.

Random-effects model (GLMM)

In parallel, a generalized linear mixed model with a logit link was fitted, including a subject-specific random intercept and a random slope for time. This structure allows for heterogeneity between patients in both baseline risk and longitudinal evolution. A simpler random-intercept-only model proved inadequate, yielding a near-zero variance estimate and numerical instability, indicating that between-subject variability could not be captured by an intercept shift alone.

Conditional mean model (subject-specific)

$$\begin{aligned} \pi_{ij} &= \Pr(Y_{ij} = 1 \mid b_{0i}, b_{1i}, t_{ij}, \mathbf{x}_i), \\ \text{logit}(\pi_{ij}) &= \beta_0 + \beta_1 t_{ij} + \beta_2 t_{ij}^2 + \mathbf{x}_i^\top \boldsymbol{\beta} + b_{0i} + b_{1i} t_{ij}. \end{aligned}$$

Random-effects distribution

$$\begin{pmatrix} b_{0i} \\ b_{1i} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{D} \right), \mathbf{D} = \begin{pmatrix} \sigma_0^2 & \sigma_{01} \\ \sigma_{01} & \sigma_1^2 \end{pmatrix}$$

(i.e., an unstructured 2×2 covariance, allowing correlation between intercept and slope).

Given (b_{0i}, b_{1i}), outcomes are conditionally independent:

$$Y_{ij} \mid b_{0i}, b_{1i} \sim \text{Bernoulli}(\pi_{ij}).$$

The final GLMM converged successfully and showed substantial variability in both random effects. The variance of the random intercept reflects pronounced differences in baseline risk across patients, while the non-zero variance of the random slope indicates meaningful heterogeneity in time trends. The estimated negative covariance between intercept and slope suggests that patients with higher initial risk tend to exhibit a slower increase over time.

The fixed effects in the GLMM again confirmed a strong non-linear time effect, with a positive linear term and a negative quadratic term, corresponding to an increasing but decelerating trajectory. In contrast to the marginal analysis, BMI and baseline tau PET were statistically significant in the GLMM, indicating that these covariates primarily influence subject-specific trajectories rather than the population-average risk. Other baseline covariates were not significant at the individual level.

Consistent with theory for non-linear models, odds ratios from the GLMM were larger in magnitude than those from the GEE, reflecting the distinction between conditional (subject-specific) and marginal (population-averaged) effects.

Comparison between GEE and GLMM

Both modeling approaches lead to consistent qualitative conclusions, particularly with respect to the dominant non-linear effect of time, which emerges as the primary driver of the binary outcome.

Table 6. Estimated odds ratios (ORs) with 95% confidence intervals from the marginal GEE model and the subject-specific GLMM for the binary outcome.

Binary outcome: key results from GEE (marginal) vs GLMM (subject-specific)

Predictor	GEE OR	GEE 95% LCL	GEE 95% UCL	GEE p	GLMM OR	GLMM 95% LCL	GLMM 95% UCL	GLMM p
abpet0	1.138	0.937	1.382	0.1933	1.134	0.630	2.040	0.6758
adl	1.004	0.983	1.026	0.6822	1.008	0.730	1.392	0.9600
age	0.999	0.987	1.010	0.8431	0.995	0.809	1.223	0.9633
bmi	0.986	0.963	1.009	0.2391	0.989	0.989	0.989	<.0001
sex=0	1.045	0.938	1.164	0.4244	1.047	0.271	4.043	0.9473
sex=1	1.000	1.000	1.000	.	1.000	.	.	.
taupet0	0.929	0.384	2.249	0.8707	0.973	0.973	0.973	<.0001
time	3.342	3.028	3.688	<.0001	3.391	2.451	4.693	<.0001
time2	0.855	0.841	0.870	<.0001	0.892	0.868	0.916	<.0001

Table 6 summarizes the estimated odds ratios from the GEE and GLMM and illustrates the systematic differences between population-averaged and subject-specific effects. In line with theory for non-linear models, odds ratios from the GLMM are larger in magnitude than those from the GEE, reflecting the distinction between conditional and marginal interpretations. While most baseline covariates show limited influence at the population level, BMI and baseline tau PET emerge as significant only in the GLMM, highlighting their role in explaining inter-individual heterogeneity rather than average population trends. The consistency of the time effects across both models strengthens confidence in the robustness of this finding.

Conclusion

In summary, the GEE and GLMM approaches provide complementary perspectives on the binary outcome. The GEE is appropriate for inference on population-averaged dynamics and is robust to misspecification of the working correlation structure, whereas the GLMM offers richer insight into subject-specific trajectories and heterogeneity. The agreement between both approaches supports the main conclusions, while the GLMM adds clinically relevant individual-level information not captured by the marginal model alone.